

MATH 4720/MSSC 5720 Spring 2026 Quiz 2

Feb 26, 2026

Name: _____ Student ID: _____

1. **Normal distribution:** Head lengths of brushtail possums follow a normal distribution with mean $\mu = 100$ mm and standard deviation $\sigma = 4$ mm, i.e., $N(100, 4^2)$.
 - (a) Calculate the z -scores for possums with head lengths of 92 mm and 104 mm.
 - (b) Which observation (92 mm or 104 mm) is more unusual or less likely to happen than the other observation? Briefly explain.

Answer:

2. **Binomial Distribution:** Offspring peas have either yellow or green pods. Assume that we want to find the probability that among five offspring peas, exactly two of them have **green** pods. Also, 75% of offspring peas have **yellow** pods. If we model the probability using the binomial distribution, identify the values of

- n , the number of trials,
- x , the number of “successes”
- π , the probability of “success”

No need to calculate any probability.

Answer: